

Module 6: Build Compare Powers and Coalition Builder Dashboards

Project: Unified Military Analytics and Comparison Dashboard

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Domain: Data Visualization

1. Methodology

Module 6 focused on implementing advanced comparative and coalition-based analytical dashboards using Power BI. This module extended the functionality of the previous dashboards by enabling side-by-side country comparison and multi-country coalition analysis. The dashboards were designed to support strategic evaluation through slicer-driven and DAX-based selection logic and aggregated metrics.

Data Modeling: A structured data model was developed in Power BI using relationships between country, military assets, and economic indicators tables. Proper one-to-many relationships were defined to ensure accurate filter propagation and aggregation. The model followed a clean star-schema approach to enhance performance and scalability.

A. Compare Powers Dashboard

Objective

To enable side-by-side comparison of any two selected countries across key military and economic metrics.

Implementation Approach

1. Country Selection Using Slicers

Two independent selection mechanisms were created using Power BI slicers:

- Country A
- Country B

These slicers allow independent country selection and dynamically update all visuals. DAX measures were created to isolate and calculate metrics separately for Country A and Country B within the same dataset.

2. KPI Cards for Direct Comparison

The following metrics were displayed for both countries:

- Manpower
- Aircraft
- Navy Fleet
- Defense Budget
- Power Index Rank
- Assets per Capita

Each KPI is dynamically populated based on slicer selections and DAX measures. This allows instant performance comparison without requiring additional filtering.

3. Visual Comparison Components

The dashboard includes multiple comparison visuals:

- Submarine comparison
- Destroyer comparison
- Naval fleet comparison
- Aircraft comparison
- Tank and attack aircraft comparison
- Assets per capita comparison

Standard Power BI visuals such as Clustered Bar Charts, Donut Charts, KPI Cards, and Matrix visuals were used to ensure clarity and interactivity. Cross-filtering behavior was enabled to allow interactive exploration of data.

4. Analytical Design Logic

The layout was structured to present:

- Summary KPIs at the top
- Detailed asset comparisons in the middle
- Domain-specific breakdowns at the bottom

This ensures a clear analytical flow from high-level comparison to detailed category-level insights.

B. Coalition Builder Dashboard

Objective

To simulate the combined military strength of multiple selected countries and compare the coalition against another country or reference benchmark.

Implementation Approach

1. Multi-Country Selection

Two selection panels were implemented:

- Select Country A (Coalition Group)
- Select Country B (Comparison Country)

The coalition group allows multiple countries to be selected simultaneously through Multi select Slicer. Power BI DAX aggregation measures were created using SUM and CALCULATE functions to compute combined totals based on coalition slicer selections.

2. Aggregated Coalition Metrics

The following coalition metrics were calculated:

- Total Manpower
- Total Defense Budget
- Total Aircraft

- Total Naval Fleet
- Total Destroyers
- Total Tanks and Attack Aircraft
- Power Index Rank comparison

All coalition metrics are dynamically aggregated using DAX measures with SUM and CALCULATE functions, filtered based on coalition slicer selections.

3. Comparative Visualizations

The dashboard includes:

- Coalition of Aircraft pie chart
- Coalition of Defense Budget bar comparison
- Coalition of Naval Fleet donut chart
- Coalition of Destroyers donut chart
- Coalition of Tanks and Attack Aircraft chart
- Coalition of Fighter Aircraft chart

These visuals enable comparison between:

- Combined coalition strength
- Individual comparison country

This supports strategic scenario analysis.

4. Analytical Structure

The design follows this logic:

- Top section: Aggregated KPIs
- Middle section: Coalition distribution visuals
- Bottom section: Asset category breakdown

This provides both macro-level and micro-level coalition insights.

C. Integration of All Dashboards

Navigation Implementation

Navigation buttons were added to ensure seamless movement between:

- Quick Stats
- Nation Overview
- Compare Powers
- Coalition Builder

Navigation buttons were implemented using Power BI Page Navigation and Action settings, enabling seamless movement between report pages.

Filter Synchronization

- Slicers were synchronized across report pages using the Sync Slicers feature in Power BI. This ensured consistent country selection and filtering behavior across all dashboards. Relationships in the data model were maintained to preserve filter context and cross-filtering functionality.
- Consistent slicer and DAX logic was maintained across report pages.

This ensures analytical continuity across the entire application.

2. Insights

Compare Powers Dashboard Insights

- Significant variation exists between countries in manpower and defense budget allocation.
- Air power and naval dominance differ substantially even among high-ranked nations.
- Assets per capita provides a normalized comparison, revealing relative military density rather than raw size.
- Some countries show strong naval dominance despite moderate manpower.
- Budget size does not always directly correlate with total military assets.

Coalition Builder Dashboard Insights

- Combined military strength of multiple mid-level countries can rival or exceed a single top-ranked nation.
- Coalition defense budgets demonstrate significant cumulative increase when major economies are grouped.
- Naval and aircraft aggregation significantly shifts strategic balance.
- Coalition simulation enables evaluation of hypothetical geopolitical alliances.
- Comparative analysis highlights how strategic partnerships alter global power positioning.

3. Deliverables Achieved

Fully functional Power BI report file

- File Name: global_military_firepower_2025.pbix
- Complete interactive Power BI report
- Integrated slicers and DAX measures
- Aggregated coalition calculations
- Navigation-enabled report structure

All Four Dashboards Fully Integrated

The following dashboards are complete and connected:

1. Quick Stats
2. Nation Overview
3. Compare Powers

4. Coalition Builder

All dashboards are interactive, slicer-driven and DAX-driven, and integrated within a single analytical framework.

Module 6 Outcome

Module 6 successfully completed the advanced analytical layer of the project by implementing dynamic comparison and coalition simulation capabilities. The final Power BI report file delivers a fully integrated military analytics application that supports global overview, country profiling, bilateral comparison, and multi-country coalition analysis within a unified system.

This module represents the final stage of dashboard development and integration for the Unified Military Analytics and Comparison Dashboard project.